Next, you will learn how to create an in-memory cluster with ElastiCache that you will later use as an in-memory cache for a gaming backend.

11.1 Creating a cache cluster

In this chapter, we focus on the Redis engine because it's more flexible. You can choose which engine to use based on the features that we compared in the previous section. If there are significant differences to Memcached, we will highlight them.

11.1.1 Minimal CloudFormation template

Imagine you are developing an online game. To do so, you need to store sessions that keep the current game state of each user, as well as a highscore list. Latency is important to ensure the gamers enjoy a great gaming experience. To be able to read and write data with very little latency, you decide to use the in-memory database Redis.

First, you need to create an ElastiCache cluster using the Management Console, the CLI, or CloudFormation. You will use CloudFormation in this chapter to manage your cluster. The resource type of an ElastiCache cluster is `AWS::ElastiCache::CacheCluster`. The required properties follow:

- `Engine`—Either `redis` or `memcached`
- `CacheNodeType`—Similar to the EC2 instance type, for example, `cache.t2.micro`
- `NumCacheNodes`—1 for a single-node cluster
- `CacheSubnetGroupName`—Reference subnets of a VPC using a dedicated resource called a subnet group
- `VpcSecurityGroupIds`—The security groups you want to attach to the cluster

A minimal CloudFormation template is shown in the next listing. The first part of the template contains the ElastiCache cluster.

Listing 11.1 Minimal CloudFormation template of an ElastiCache Redis single-node cluster